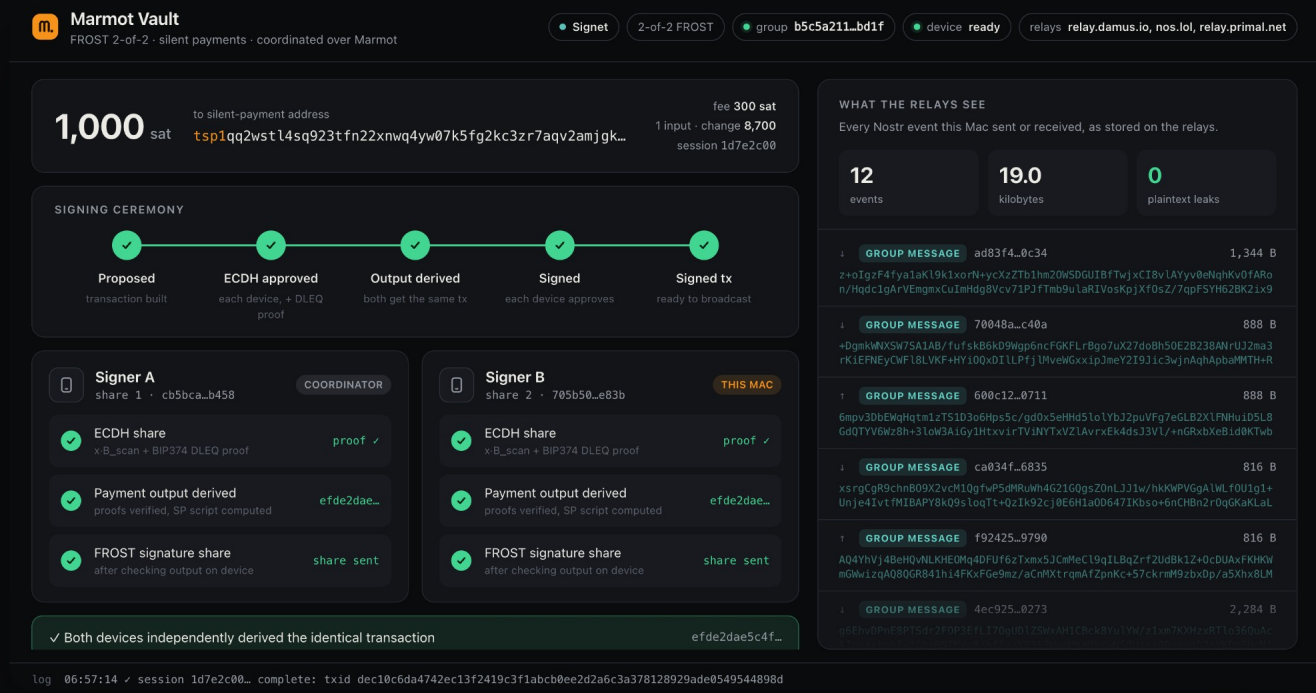
BITCOIN++ SEOUL · BEST USE OF MARMOT

# BlackIce

## Distributed Frost Silent Payments

Hardware signers on separate devices co-sign a bitcoin payment to a Silent Payment address, coordinated over an encrypted Marmot group.




BlackIce: Robos&Rademenez · Signet · November 2026

# Silent payments, now from a hardware multisig

UNTIL NOW

## Single-key wallets only

To pay a silent-payment address, the sender needs the combined secret of every input key. A multisig never has that in one place, so no multisig could send to one.

WITH BLACKICE

## Hardware multisig pays them

Each Frostsnap device contributes its own ECDH share with a proof. The shares combine into the payment output, and no key ever leaves its device.

Frostsnap hardware

FROST threshold keys, any k-of-n

Confirmed on Signet

# Three layers, each one private

MARMOT

## Encrypted coordination

MLS group messaging over Nostr. Relays store only ciphertext. No phone number, no server.

Fixes the chat

FROST 2-OF-2

## One key, one signature

Two devices sign as one Schnorr key. On-chain it’s an ordinary single-key taproot spend.

Fixes the chain

SILENT PAYMENTS

## Unlinkable outputs

One static address. Each payment lands in a fresh output that nobody else can link to it.

Fixes the recipient

Until now, only single-key wallets could pay a silent-payment address. **BlackIce sends from a hardware multisig.**

# How a payment gets signed

1

### Propose

Signer A's bridge builds the transaction. The payment output can't exist yet.

2

### ECDH + proof

Each device asks its owner, then returns its ECDH share with a BIP374 DLEQ proof.

3

### Derive

Each bridge verifies both proofs and computes the payment output. Both results must match.

4

### Sign

Each device shows the real output. Its owner approves and it returns a FROST share.

5

### Combine

Two shares become one Schnorr signature, ready to broadcast.

Every message between steps is an MLS-encrypted Marmot message on public Nostr relays.

LIVE ON SIGNET

# Two devices, two laptops, one encrypted group

STEP 7 / 7

## One Schnorr signature. 0 plaintext leaks on the relays

**Marmot Vault**
FROST 2-of-2 · silent payments · coordinated over Marmot

Signet · 2-of-2 FROST · group b5c5a211…bd1f · device ready · relays relay.damus.io, nos.lol, relay.primal.net

**1,000** sat
to silent-payment address
tsp1qq2wstl4sq923tfn22xnwq4yw07k5fg2kc3zr7aqv2amjgk…
fee 300 sat
1 input · change 8,700
session 1d7e2c00

SIGNING CEREMONY

| Proposed | ECDH approved | Output derived | Signed | Signed tx |
|---|---|---|---|---|
| transaction built | each device, + DLEQ proof | both get the same tx | each device approves | ready to broadcast |

**Signer A** — share 1 · cb5bca…b458 — COORDINATOR

| | | |
|---|---|---|
| ECDH share | x·B_scan + BIP374 DLEQ proof | proof ✓ |
| Payment output derived | proofs verified, SP script computed | efde2dae… |
| FROST signature share | after checking output on device | share sent |

**Signer B** — share 2 · 705b50…e83b — THIS MAC

| | | |
|---|---|---|
| ECDH share | x·B_scan + BIP374 DLEQ proof | proof ✓ |
| Payment output derived | proofs verified, SP script computed | efde2dae… |
| FROST signature share | after checking output on device | share sent |

✓ Both devices independently derived the identical transaction — efde2dae5c4f…

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

| 12 | 19.0 | 0 |
|---|---|---|
| events | kilobytes | plaintext leaks |

| | | |
|---|---|---|
| ↓ GROUP MESSAGE | ad83f4…0c34 | 1,344 B |
| | z+oIgzF4fya1aKl9k1xorN+ycXzZTb1hm2OWSDGUIBfTwjxCI8vlAYyv0eNqhKvOfARo n/Hqdc1gArVEmgmxCuImHdg8Vcv71PJfTmb9ulaRIVosKpjXfOsZ/7qpF5YH62BK2ix9 | |
| ↓ GROUP MESSAGE | 70048a…c40a | 888 B |
| | +DgmkWNXSW7SA1AB/fufskB6kD9Wgp6ncFGKFLrBgo7uX27doBh5OE2B238ANrUJ2ma3 rKiEFNEyCWFl8LVKF+HYiOQxDIlLPfjlMveWGxxipJmeY2I9Jic3wjnAqhApbaMMTH+R | |
| ↑ GROUP MESSAGE | 600c12…0711 | 888 B |
| | 6mpv3DbEWqHqtm1zTS1D3o6Hps5c/gdOx5eHHd5lolYbJ2puVFg7eGLB2XlFNHuiD5L8 GdQTYV6Wz8h+3loW3AiGy1HtxvirTViNYTxVZlAvrxEk4dsJ3Vl/+nGRxbXeBid0KTwb | |
| ↓ GROUP MESSAGE | ca034f…6835 | 816 B |
| | xsrgCgR9chnBO9X2vcM1QgfwP5dMRuWh4G21GQgsZOnLJJ1w/hkKWPVGgAlWLfOU1g1+ Unje4IvtfMIBAPY8kQ9sloqTt+QzIk92cj0E6H1aOD647IKbso+6nCHBn2rOqGKaKLaL | |
| ↑ GROUP MESSAGE | f92425…9790 | 816 B |
| | AQ4YhVj4BeHQvNLKHEOMq4DFUf6zTxmx5JCmMeCl9qILBqZrf2UdBk1Z+OcDUAxFKHKW mGWwizqAQ8QGR841hi4FKxFGe9mz/aCnMXtrqmAfZpnKc+57ckrmM9zbxDp/a5Xhx8LM | |
| ↓ GROUP MESSAGE | 4ec925…0273 | 2,284 B |
| | g6EhvDPnE8PTSdr2FOP3EfLI7OqUDlZSWxAH1CBck8YulYW/z1xm7KXHzxRT1o36QuAc | |

log 06:57:14 ✓ session 1d7e2c00… complete: txid dec10c6da4742ec13f2419c3f1abcb0ee2d2a6c3a378128929ade0549544898d

# The device checks it, not the laptop




STEP 2 · ECDH

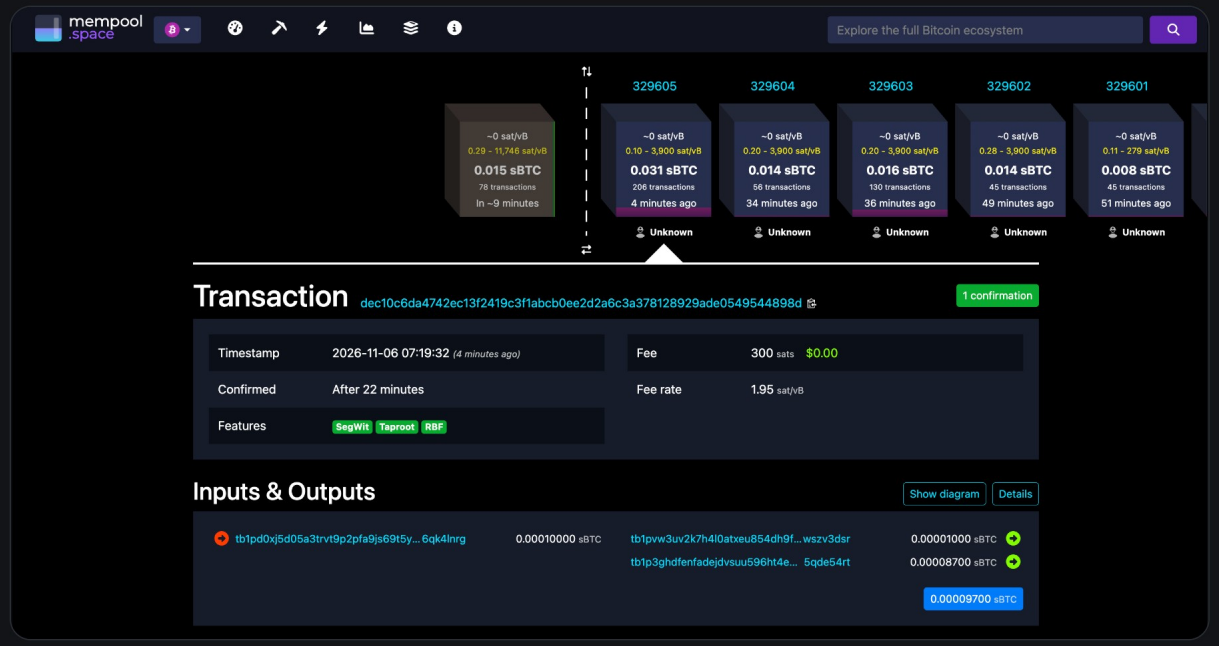
“Hold to Contribute” adds the device’s key share. It isn’t signing yet.




STEP 4 · SIGN

The device shows the silent-payment address, amount and fee, then “Hold to Sign”.

# On-chain, it’s just a taproot spend




**CONFIRMED · BLOCK 329,605**

`dec10c6da4742ec13f2419c3`
`f1abcb0ee2d2a6c3a3781289`
`29ade0549544898d`

**≈154 vB**: one key-path input and two taproot outputs. No script, no second key, no multisig fingerprint.

The 1,000 sat landed at tb1pvw3u…v3dsr, a fresh taproot address nobody can link to the recipient’s tsp1… address.

mempool.space/signet · confirmed 2026-11-06 07:19 · fee 300 sat at 1.95 sat/vB

# What’s new, and what’s next

### BUILT THIS WEEKEND

- Remote FROST signing on real hardware, coordinated only over Marmot
- A hardware multisig that pays silent-payment addresses, with per-device DLEQ proofs (Frostsnap PR #640)
- No coordinator server; joining the group grants no signing power
- No signature share leaves a device until both derive the identical transaction

### NEXT

- Import transactions from Sparrow
- k-of-n vaults, such as 2-of-3
- Receive to the vault’s own silent-payment address
- Fix the device’s intermittent serial crash upstream

BlackIce

# Distributed. Frost. Silent Payments.

No phone number. No server. No on-chain fingerprint.

`github.com/rademenezbtc/blackice`